Boss Challenge — Paper 37 (Class 7)

Forests | Soil | Integers | Arithmetic Expressions

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The uppermost layer of a forest, formed by the spreading branches of the tallest trees, is called the:

(A) bedrock
(B) understorey
(C) canopy
(D) forest floor

2. Positive numbers, negative numbers and zero, such as -3, 0 and +5, are together called:

(A) fractions
(B) multiples
(C) integers
(D) decimals

3. The slow breaking down of rocks into fine particles over a very long time, which forms soil, is called:

(A) germination
(B) evaporation
(C) erosion
(D) weathering

4. In the expression 7 + 4, the numbers 7 and 4 that are joined by the plus sign are called the:

(A) products
(B) brackets
(C) terms
(D) factors

5. The branchy top part of a single tree, made of all its branches and leaves, is called its:

(A) trunk
(B) canopy
(C) crown
(D) root

6. On a number line, the integers lying to the left of zero, such as -2 and -5, are all:

(A) negative
(B) positive
(C) zero
(D) fractions

7. A vertical cut showing the different layers of soil from the surface down to solid rock is called the:

(A) food chain
(B) canopy
(C) soil profile
(D) water table

8. The result obtained by adding numbers together is called their:

(A) sum
(B) quotient
(C) difference
(D) product

9. Below the tallest trees, the shorter trees and large shrubs that receive only partial sunlight form the:

(A) subsoil
(B) humus
(C) understorey
(D) canopy

10. The one integer that is neither positive nor negative is:

(A) ten
(B) one
(C) zero
(D) -1

11. A single distinct band within the soil profile, such as the dark topsoil, is known as a:

(A) orbit
(B) term
(C) horizon
(D) crown

12. The result obtained by multiplying numbers together is called their:

(A) product
(B) sum
(C) quotient
(D) difference

13. The dark, crumbly substance formed in the topsoil from rotted dead leaves and animal remains is:

(A) sand
(B) humus
(C) clay
(D) bedrock

14. A seed is buried 4 cm below the soil surface, shown as the integer -4. If it is lifted 4 cm up to the surface, it reaches the integer:

- **(A)** -4
- **(B)** 0
- **(C)** -8
- **(D)** 4

15. The topmost, dark soil layer that is richest in humus and where most plant roots grow is called the:

- **(A)** topsoil
- **(B)** canopy
- **(C)** bedrock
- **(D)** subsoil

16. In the expression 6 x 3, the result of the multiplication, that is the product, is:

- **(A)** 18
- **(B)** 63
- **(C)** 3
- **(D)** 9

17. Tiny organisms such as bacteria and fungi that break down dead leaves and dead animals into humus are called:

- **(A)** herbivores
- **(B)** decomposers
- **(C)** predators
- **(D)** producers

18. The sum -3 + 5 equals:

- **(A)** 8
- **(B)** -8
- **(C)** 2
- **(D)** -2

19. The layer of solid, unbroken rock lying at the very bottom of a soil profile is the:

- **(A)** subsoil
- **(B)** bedrock
- **(C)** topsoil
- **(D)** humus

20. To evaluate 4 + 5 x 2, the multiplication is done first, giving:

- **(A)** 13
- **(B)** 18
- **(C)** 14
- **(D)** 40

21. Forests are often called the 'green lungs' of the Earth because, in daylight, their trees give out:

- **(A)** oxygen
- **(B)** nitrogen
- **(C)** carbon dioxide
- **(D)** smoke

22. The sum -6 + (-2) equals:

- **(A)** -8
- **(B)** 4
- **(C)** -4
- **(D)** 8

23. Soil made of the largest particles, through which water drains away very quickly, is:

- **(A)** clayey soil
- **(B)** sandy soil
- **(C)** loamy soil
- **(D)** humus

24. In the expression 12 - 2 x 3, doing the multiplication first gives:

- **(A)** 6
- **(B)** 8
- **(C)** 36
- **(D)** 30

25. Trees help balance the air by taking in the gas that animals breathe out; that gas is:

- **(A)** hydrogen
- **(B)** oxygen
- **(C)** water vapour
- **(D)** carbon dioxide

26. The difference 4 - 9 equals:

- **(A)** -5
- **(B)** -13
- **(C)** 5
- **(D)** 13

27. Soil made of very fine, tightly packed particles, which holds a great deal of water, is:

- **(A)** clayey soil
- **(B)** sandy soil
- **(C)** loamy soil
- **(D)** bedrock

28. Because the bracket is worked first, the expression (3 + 4) x 2 equals:

- **(A)** 11
- **(B)** 24
- **(C)** 9
- **(D)** 14

29. Fallen leaves and tree roots let rainwater soak slowly into the ground rather than run off, which helps to:

- **(A)** pollute the soil
- **(B)** cause floods
- **(C)** dry up the rivers
- **(D)** recharge groundwater

30. At night a forest is at -2°C; by noon it rises by 7°C. The noon temperature, found by -2 + 7, is:

- **(A)** -5°C
- **(B)** 9°C
- **(C)** 5°C
- **(D)** -9°C

31. Crops grow best in a balanced mix of sand, silt and clay; this all-round soil is called:

- **(A)** clayey soil
- **(B)** sandy soil
- **(C)** bedrock
- **(D)** loamy soil

32. Without any bracket, the expression 3 + 4 x 2 equals:

- **(A)** 24
- **(B)** 9
- **(C)** 11
- **(D)** 14

33. By gripping the soil with their roots, forest trees prevent the fertile topsoil from being washed away - that is, they prevent:

- **(A)** soil erosion
- **(B)** germination
- **(C)** evaporation
- **(D)** photosynthesis

34. The product (-3) x 4 equals:

- **(A)** 12
- **(B)** -12
- **(C)** -7
- **(D)** -1

35. The trickling of water slowly downward through the soil is called:

- **(A)** transpiration
- **(B)** evaporation
- **(C)** percolation
- **(D)** condensation

36. Comparing (3 + 4) x 2 with 3 + 4 x 2, the two expressions are:

(A) always equal
(B) both equal to 24
(C) not equal
(D) both equal to 9

37. The sequence that shows who eats whom, such as grass -> deer -> tiger, is called a:

(A) life cycle
(B) food web
(C) water cycle
(D) food chain

38. The product (-5) x (-2) equals:

(A) -10
(B) -7
(C) 7
(D) 10

39. The amount of water that a particular soil is able to hold is called its:

(A) percolation rate
(B) weathering
(C) temperature
(D) water-holding capacity

40. The value of 20 / (2 + 3) is:

(A) 25
(B) 5
(C) 4
(D) 13

41. When many food chains in a forest cross and link with one another, together they form a:

(A) canopy
(B) crown
(C) food web
(D) food chain

42. Dividing (-12) / 3 gives:

(A) -4
(B) -36
(C) 4
(D) -9

43. Sandy soil has a low water-holding capacity mainly because water passes through its large gaps:

(A) upward
(B) quickly
(C) slowly
(D) never

44. The value of 2 x (5 + 1) is:

(A) 12
(B) 16
(C) 8
(D) 11

45. Forest animals such as deer and rabbits, which eat only plants, are called:

(A) decomposers
(B) herbivores
(C) carnivores
(D) producers

46. Dividing (-20) / (-4) gives:

(A) -16
(B) 5
(C) -5
(D) 80

47. Clayey soil is well suited to growing paddy (rice) because clayey soil is able to:

(A) stop roots growing
(B) hold water well
(C) make its own water
(D) drain water fast

48. The expression 5 x 8 has the same value as 8 x 5 because multiplication can be done in any:

(A) size only
(B) colour
(C) direction
(D) order

49. Forest animals such as the tiger, which hunt and eat other animals, are called:

(A) decomposers
(B) herbivores
(C) carnivores
(D) producers

50. The additive inverse (the opposite) of +7 is:

- **(A)** +7
- **(B)** -7
- **(C)** 0
- **(D)** 1/7

51. The mixing of harmful substances such as plastic and chemicals into the soil is called:

- **(A)** soil pollution
- **(B)** percolation
- **(C)** weathering
- **(D)** soil erosion

52. A forest patch is laid out as 4 plots, each holding (10 trees + 2 saplings). The total number of plants, found by 4 x (10 + 2), is:

- **(A)** 42
- **(B)** 18
- **(C)** 16
- **(D)** 48

53. The cutting down of forests on a large scale for timber, farming or building is called:

- **(A)** afforestation
- **(B)** decomposition
- **(C)** deforestation
- **(D)** germination

54. On a number line, the greater of the two integers -1 and -6 is:

- **(A)** -6
- **(B)** they are equal
- **(C)** 0
- **(D)** -1

55. The carrying away of the fertile topsoil by flowing water and strong wind is called:

- **(A)** percolation
- **(B)** soil pollution
- **(C)** weathering
- **(D)** soil erosion

56. Three bags of soil weigh 25 kg each, and an extra 10 kg is added on top. The total mass, found by 3 x 25 + 10, is:

- **(A)** 105 kg
- **(B)** 85 kg
- **(C)** 75 kg
- **(D)** 38 kg

57. Seeds of many forest plants are carried to new places by wind, water and animals; this scattering of seeds is called:

- **(A)** pollination
- **(B)** germination
- **(C)** seed dispersal
- **(D)** decomposition

58. An earthworm is at a depth of -15 cm and burrows 5 cm deeper. Its new depth, found by -15 + (-5), is:

- **(A)** 20 cm
- **(B)** -75 cm
- **(C)** -10 cm
- **(D)** -20 cm

59. The rotted remains of plants and animals that make the topsoil dark and fertile are the:

- **(A)** humus
- **(B)** bedrock
- **(C)** clay
- **(D)** sand

60. The value of 100 - (40 + 30) is:

- **(A)** 170
- **(B)** 30
- **(C)** 70
- **(D)** 90

61. Fallen trees in a forest are slowly replaced as new plants grow on their own, so the forest renews itself; this natural renewal is called:

- **(A)** erosion
- **(B)** regeneration
- **(C)** deforestation
- **(D)** evaporation

62. The absolute value |-8|, which is the distance of -8 from zero, equals:

- **(A)** 16
- **(B)** 0
- **(C)** -8
- **(D)** 8

63. A soil scientist marks the topsoil as ending at -20 cm and the subsoil as ending at -50 cm below ground. The thickness of the subsoil layer, found by the gap between 20 cm and 50 cm, is:

- **(A)** 30 cm
- **(B)** 50 cm
- **(C)** 70 cm
- **(D)** 20 cm

64. The repeated sum 6 + 6 + 6 can be written more shortly as the product:

- **(A)** 3 + 6
- **(B)** 3 x 6
- **(C)** 6 + 3
- **(D)** 6 x 6

65. The smallest, soft green plants growing on the forest floor, such as grasses and small flowering plants, are the:

- **(A)** herbs
- **(B)** trees
- **(C)** canopy
- **(D)** shrubs

66. Adding any integer to zero, for example -9 + 0, gives back:

- **(A)** -9
- **(B)** 9
- **(C)** -18
- **(D)** 0

67. The lighter-coloured layer that lies just below the topsoil, with less humus and more small rock pieces, is the:

- **(A)** bedrock
- **(B)** canopy
- **(C)** topsoil
- **(D)** subsoil

68. Using the distributive idea, 6 x (10 + 1) is the same as 6 x 10 + 6 x 1, which equals:

- **(A)** 61
- **(B)** 60
- **(C)** 17
- **(D)** 66

69. Medium-sized woody plants, larger than the floor herbs but shorter than trees, are the:

- **(A)** shrubs
- **(B)** herbs
- **(C)** trees
- **(D)** decomposers

70. The product of any integer and zero, for example (-7) x 0, is always:

- **(A)** -70
- **(B)** 0
- **(C)** 7
- **(D)** -7

71. When wet, clayey soil can be rolled into a ball because it feels:

(A) hard like rock
(B) dry and dusty
(C) smooth and sticky
(D) rough and gritty

72. A forester plants 5 rows of 8 trees, but 4 trees later fall. The number left, found by 5 x 8 - 4, is:

(A) 36
(B) 40
(C) 20
(D) 9

73. The government department whose job is to look after, manage and protect the forests is the:

(A) Health Department
(B) Forest Department
(C) Police Department
(D) Water Department

74. The sum of an integer and its opposite, for example 6 + (-6), is:

(A) 6
(B) 0
(C) 12
(D) -12

75. When you rub dry sandy soil between your fingers, it feels:

(A) rough and gritty
(B) wet and slippery
(C) soft like flour
(D) smooth and sticky

76. Adding zero to a number, as in 15 + 0, leaves the value as:

(A) 16
(B) 0
(C) 150
(D) 15

77. When decomposers break down dead leaves, the nutrients locked inside are returned back to the:

(A) clouds
(B) air
(C) soil
(D) sunlight

78. A fish floats at -3 m and a stone rests at -9 m below a forest pond's surface. The difference in their depths, found by -3 - (-9), is:

- **(A)** 12 m
- **(B)** -12 m
- **(C)** 3 m
- **(D)** 6 m

79. Apart from anchoring their roots, plants depend on the soil to supply them with water and dissolved:

- **(A)** oxygen for photosynthesis
- **(B)** minerals
- **(C)** sunlight
- **(D)** plastic

80. The value of 24 / 6 + 1 is:

- **(A)** 5
- **(B)** 4
- **(C)** 25
- **(D)** 3

81. Forests help bring rain because water vapour rises into the air from the leaves of trees through the process of:

- **(A)** condensation
- **(B)** transpiration
- **(C)** germination
- **(D)** respiration

82. The product (-1) x (-1) x (-1) equals:

- **(A)** 1
- **(B)** -1
- **(C)** 3
- **(D)** -3

83. A root tip is growing at a depth of -12 cm and pushes 6 cm deeper into the soil. Its new depth, found by -12 + (-6), is:

- **(A)** -18 cm
- **(B)** 18 cm
- **(C)** -6 cm
- **(D)** -72 cm

84. The value of 2 x 2 x 2 is:

- **(A)** 6
- **(B)** 2
- **(C)** 4
- **(D)** 8

85. A small forest patch is planted in 8 rows with 9 trees in each row. The total number of trees, found by 8 x 9, is:

- **(A)** 1
- **(B)** 17
- **(C)** 81
- **(D)** 72

86. Among the integers -4, 0, 3 and -7, the smallest is:

- **(A)** 0
- **(B)** -4
- **(C)** 3
- **(D)** -7

87. Earthworms are called a farmer's friend because, by burrowing through the ground, they make the soil more:

- **(A)** salty
- **(B)** loose and airy
- **(C)** hard and packed
- **(D)** waterproof

88. To make the value of 2 + 3 x 4 come out as 20, you must place a bracket to get:

- **(A)** 2 x 3 + 4
- **(B)** (2 + 3) x 4
- **(C)** 2 + (3 x 4)
- **(D)** (2 + 3) + 4

89. In a tree census a forester counts 40 trees and finds that 5 of them are dead. The number of living trees, found by 40 - 5, is:

- **(A)** 35
- **(B)** 8
- **(C)** 45
- **(D)** 200

90. Of two integers shown on a number line, the one lying farther to the right is always the:

- **(A)** smaller
- **(B)** negative
- **(C)** equal
- **(D)** greater

91. Using far too much chemical fertilizer year after year can, over time, damage the soil's:

- **(A)** colour only
- **(B)** shape
- **(C)** fertility
- **(D)** temperature

92. Comparing the two products 5 x 10 and 5 x 9, the larger one is:

- **(A)** neither
- **(B)** 5 x 9
- **(C)** they are equal
- **(D)** 5 x 10

93. Green plants in a forest, which make their own food using sunlight and stand at the start of every food chain, are called:

- **(A)** consumers
- **(B)** decomposers
- **(C)** producers
- **(D)** predators

94. A hill forest is at 3°C and the temperature drops by 8°C overnight. The new temperature, found by 3 - 8, is:

- **(A)** -11°C
- **(B)** 5°C
- **(C)** 11°C
- **(D)** -5°C

95. A field is divided into 6 equal plots, and each plot needs 7 kg of compost. The total compost required, found by 6 x 7, is:

- **(A)** 1 kg
- **(B)** 36 kg
- **(C)** 42 kg
- **(D)** 13 kg

96. A topsoil sample gains 3 g of humus each year for 7 years, on top of a starting 4 g. Its total humus, found by 3 x 7 + 4, is:

- **(A)** 49 g
- **(B)** 14 g
- **(C)** 21 g
- **(D)** 25 g

97. The whole variety of different plants and animals living together in a forest is called its:

- **(A)** erosion
- **(B)** biodiversity
- **(C)** humus
- **(D)** canopy

98. Subtracting a negative number, as in 5 - (-3), gives:

- **(A)** 2
- **(B)** -8
- **(C)** 8
- **(D)** -2

99. Crops like wheat and gram grow best in soil that holds both water and air well, namely:

- **(A)** pure clayey soil
- **(B)** loamy soil
- **(C)** bedrock
- **(D)** pure sandy soil

100. Worked from left to right, the expression 8 - 3 + 2 equals:

- **(A)** 7
- **(B)** 3
- **(C)** 13
- **(D)** 9